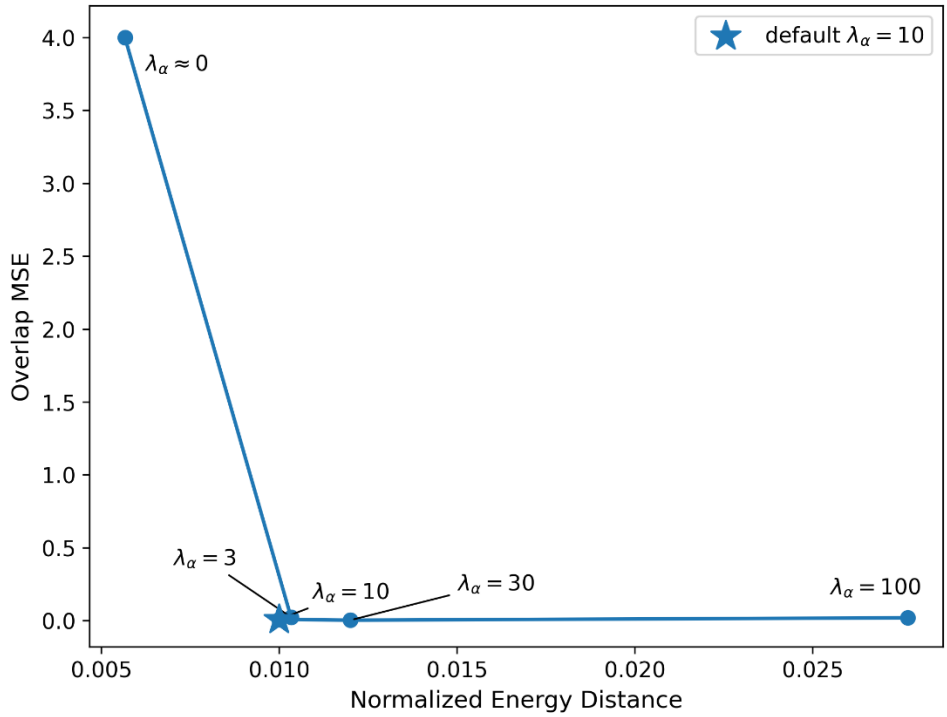

Figure 1. Realism–overlap trade-off under overlap rigidity (λ_α) sweep. Normalized energy distance measures global distributional fidelity (lower is better). Increasing $\lambda_\alpha$substantially improves overlap calibration (lower MSE), with only a modest increase in distributional discrepancy, revealing a stable operating region around $\lambda_\alpha \approx 10$–30.

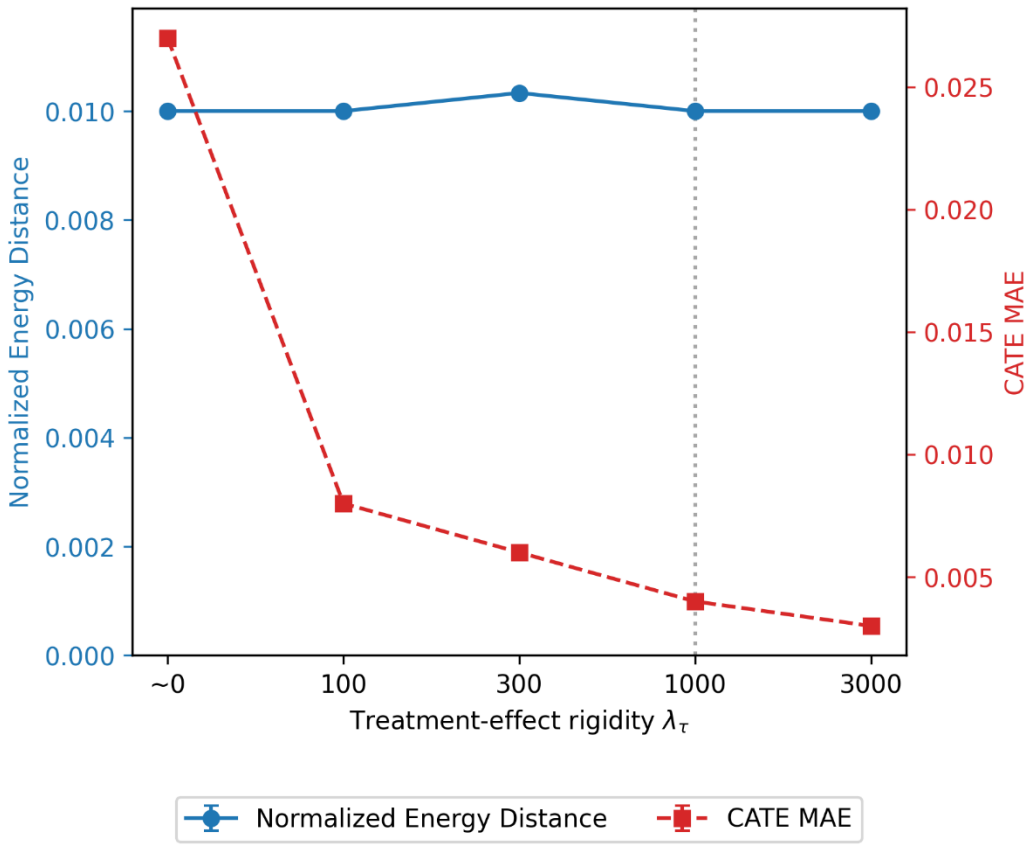

Figure 2. Realism–treatment effect trade-off under treatment-effect rigidity (λ_τ) sweep. Normalized energy distance measures global distributional fidelity (lower is better). Increasing $\lambda_\tau$substantially improves treatment-effect alignment (lower CATE MAE) while maintaining nearly constant distributional realism, indicating a broad stable operating region around the default λ_τ of 10^3.

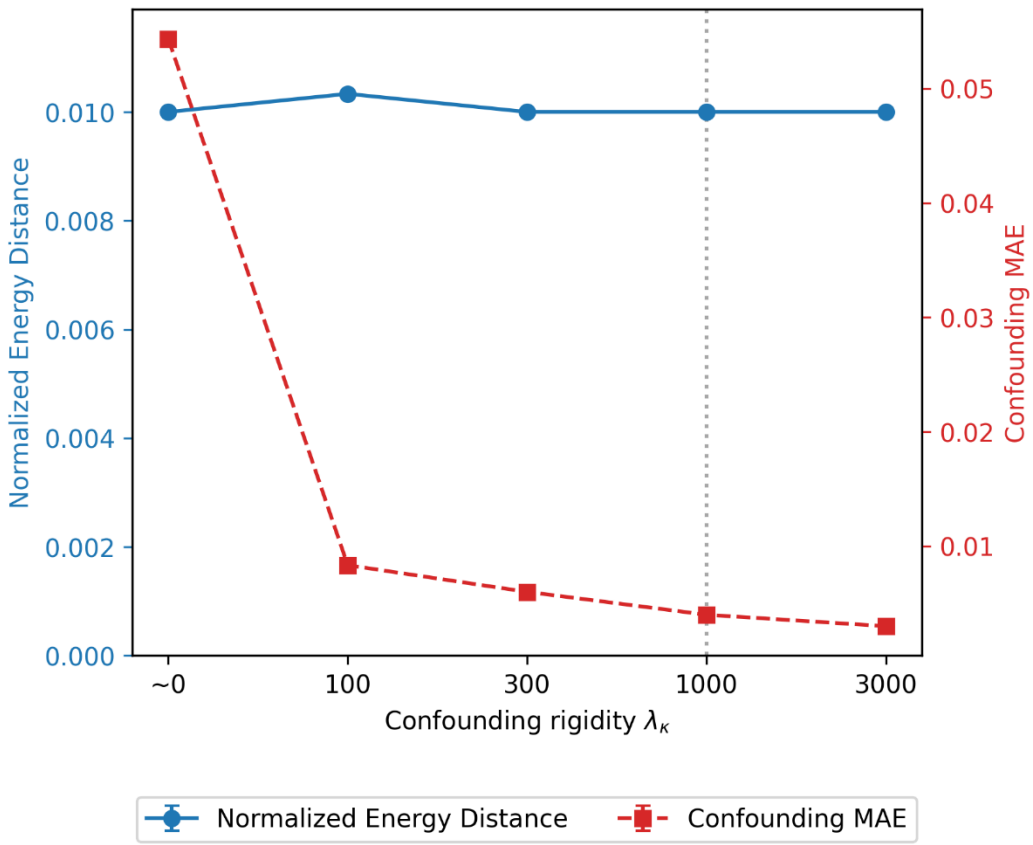

Figure 3. Realism–confounding trade-off under confounding rigidity (λ_κ) sweep. Normalized energy distance measures global distributional fidelity (lower is better). Increasing $\lambda_\kappa$substantially improves confounding alignment (lower MAE) while maintaining nearly constant distributional realism, indicating a stable operating region around the default λ_κ of 10^3.